Biomedical Foundation Models

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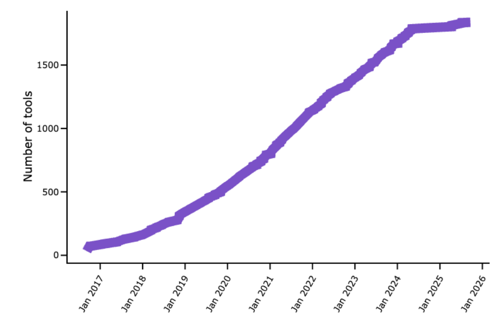
Do we have a single, best solution to these questions?

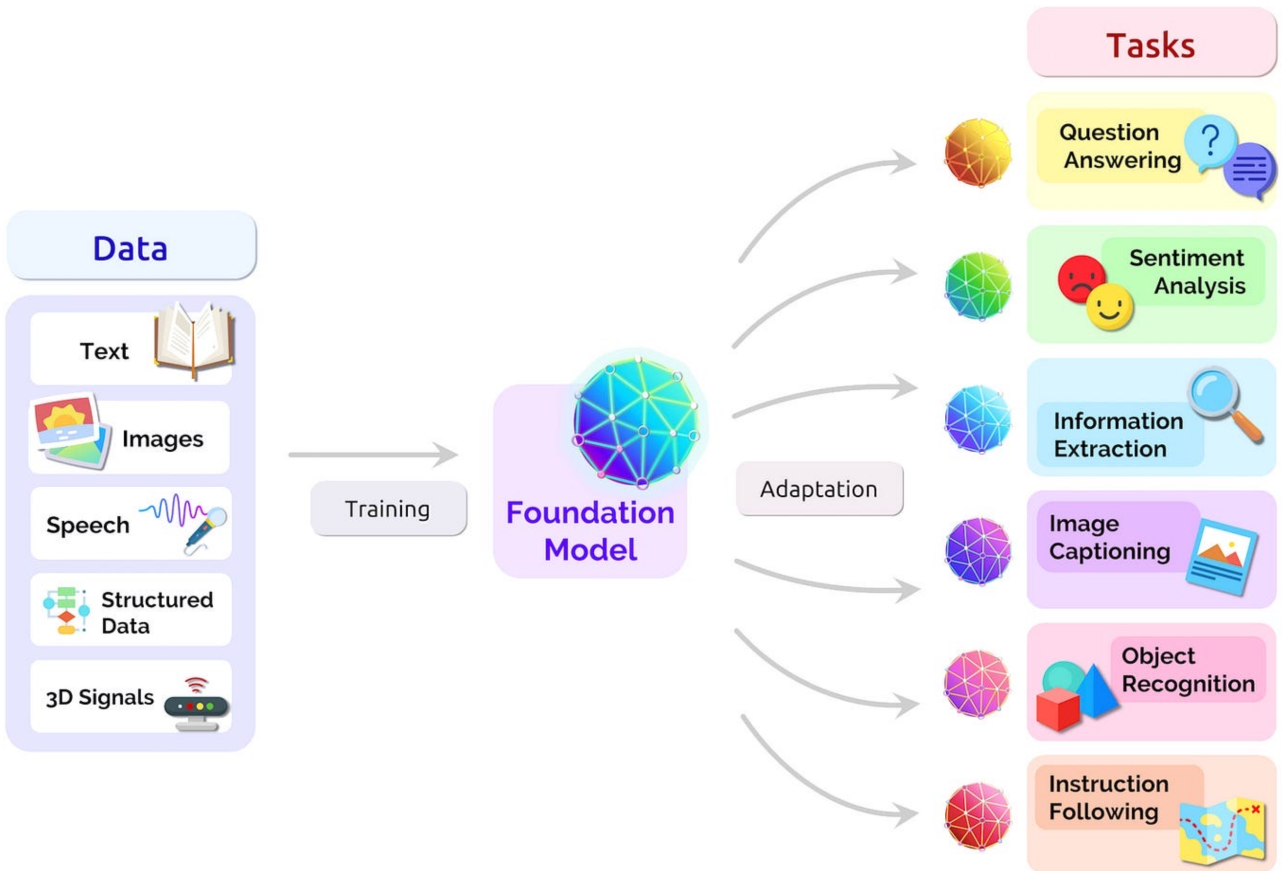
- **How do I get directions from location A to B?**
- **How to carry out Differential Expressed Gene (DEG) analysis?**
- **How to carry out single-cell clustering analysis?**
- **How to create a cancer-free world?**

- **Belief Theory**, also referred to as **evidence theory** or **Dempster–Shafer theory** (**DST**), is a general framework for reasoning with uncertainty.
- The theory allows one to combine evidence from different sources and arrive at a degree of belief (**good-enough solutions**) that takes into account all the available evidence.



We currently track 1837 tools...

The concept of foundation models



- Trained on **vast and diverse** datasets (e.g., text, images, speech, structured data, 3D signals)
- A large-scale deep learning model with **a lot** of parameters
- Learn broad, general representations that can be adapted to **various downstream tasks**.

Bommasani et al, 2022, https://arxiv.org/pdf/2108.07258.pdf

PriMed AI: Integrating Imaging with Multimodal Data (2026)

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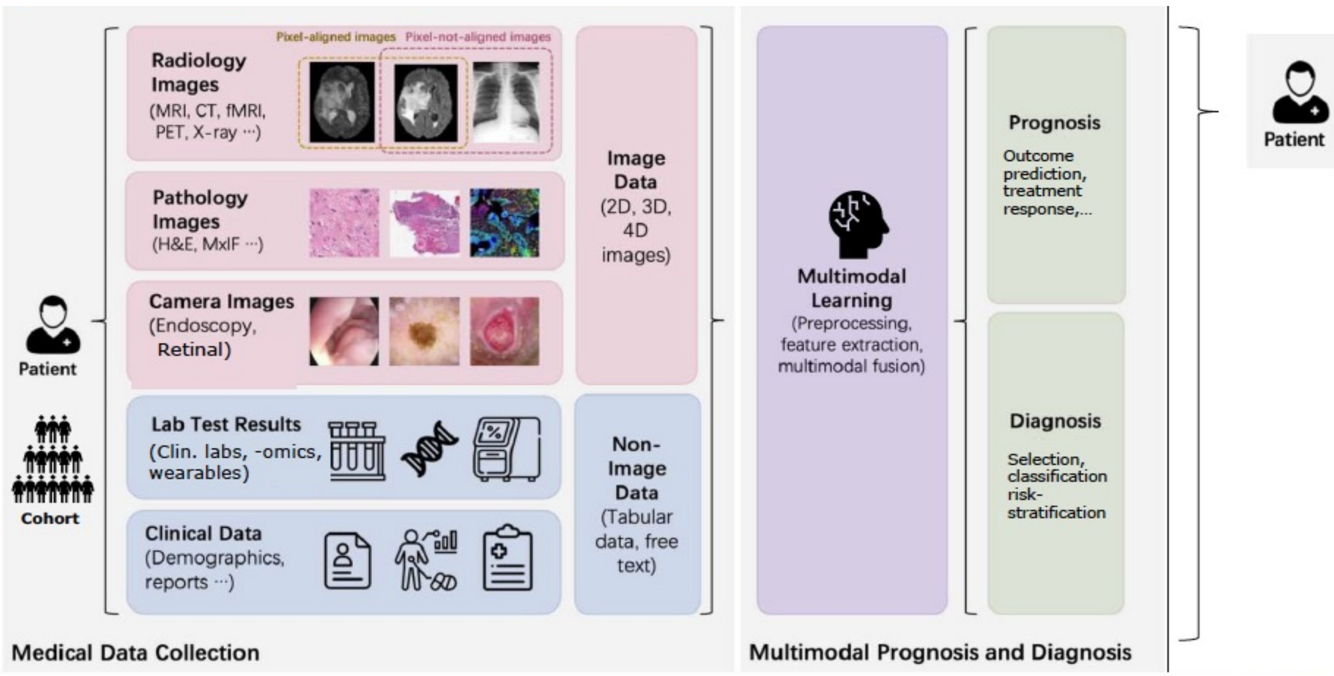


To catalyze the development & adoption of innovative **AI-based clinical decision support tools** that **integrate clinical imaging with multimodal health data** to enable reliable, cost-effective, accessible, and sustainable **precision medicine** workflows for diagnosis, treatment, and quality of care.

Program Duration: 5 years

Funds Available and Anticipated Number of Awards: ~25M per year (FY27-31), ~90 meritorious awards (over five years, contingent upon availability of funds)

Council Approved: April 2025

Est RFA release date: April 1, 2026

Est submission due date: June 1, 2026

Est project start date: March 1, 2027

NCI: Foundation models for Cancer Research (March 2026)

- **Foundation Model Primer**
- **Multimodal Data:** Combining pathology, radiology, omics, and patient data into unified models.
- **Prediction:** therapeutic response, resistance, and patient outcomes.
- **Validation and Reproducibility:** real-world clinical performance and use.
- **Diagnostic Case Studies:** Real-world applications for early detection and automated diagnostics.
- **Federated Learning:** training robust models across multiple institutions without sharing sensitive patient data
- **Challenges, Risk, and Clinical Use:** Addressing model interpretability and considerations for adoption and deployment in real-world clinical settings.

NIH WORKSHOP

Foundation Models for Cancer

Advancing Diagnosis, Prognosis, and Treatment Response

March 24-26, 2026
10:00 AM - 2:00 PM ET
Virtual

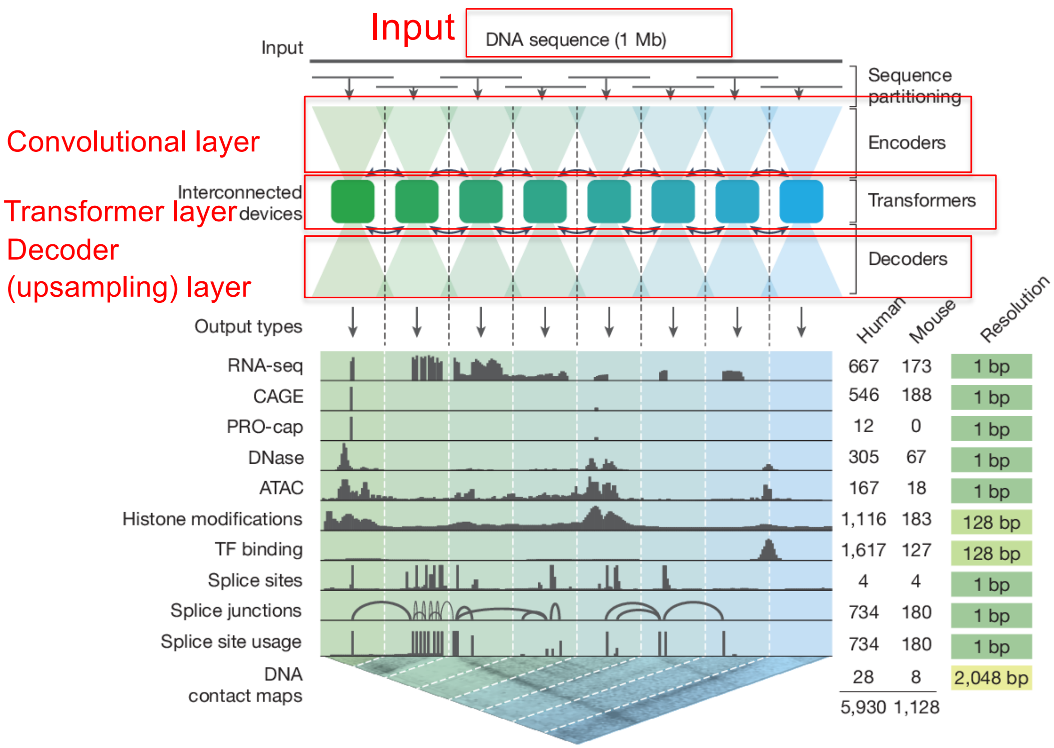
Zuckerberg and Chan dropped $500M to build the first real “Google Maps” for human cells (April 2026)

Funding Split: $400 million will be used for generating new data and next-generation imaging technology, while $100 million will fund external research labs, including collaborations with institutions like the **Allen Institute** and **NVIDIA**.

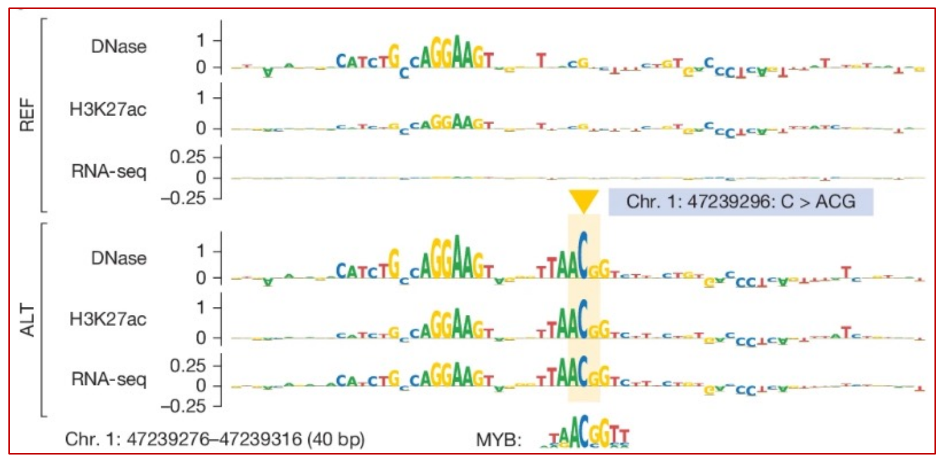
"Virtual Cells" Goal: The objective is to create **AI models** that can simulate the behavior of healthy and diseased cells, allowing researchers to see inside cells at unprecedented speeds and detail.

Addressing Disease: This initiative is aimed at helping researchers understand disease mechanisms and develop personalized treatments, moving beyond trial-and-error medicine.

AlphaGenome: Foundation model trained on human genomics

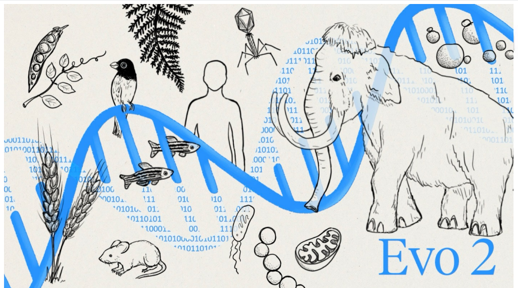


Google DeepMind, Nature 2026

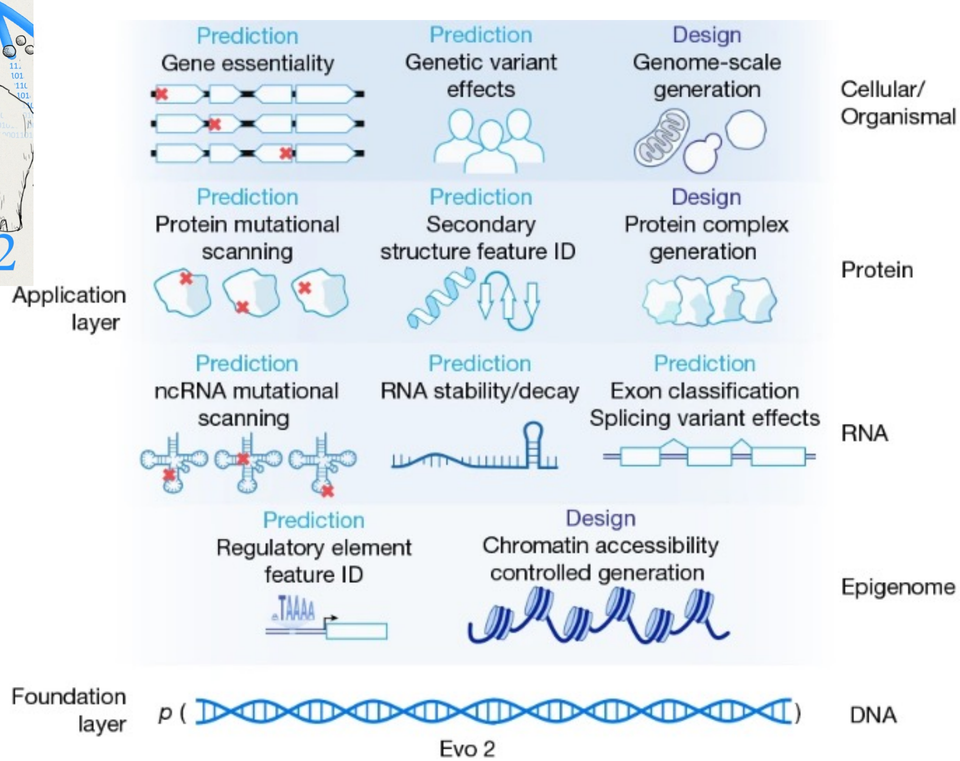


Computational assays outputs:
11 biological modalities predicted simultaneously

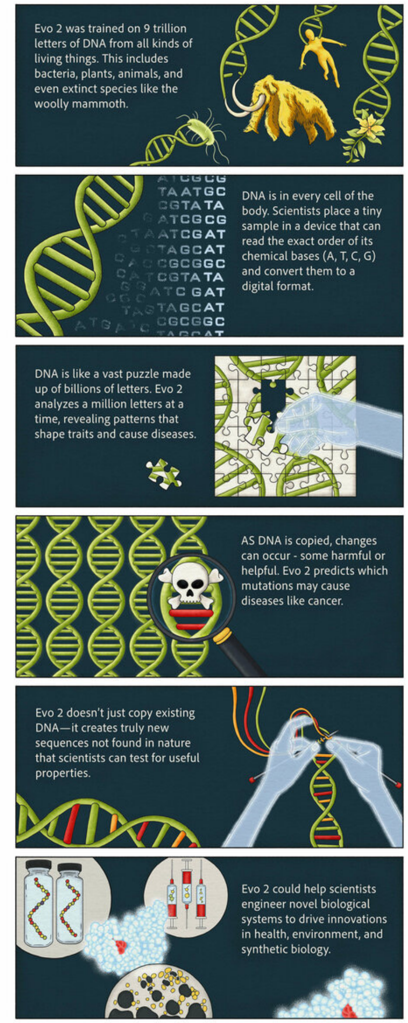
Evo2: Foundation model trained on 100,000+ genomes



Stanford Team
Nature 2026



Evo 2 models DNA sequence and enables applications across the central dogma, scaling from molecules to genomes and spanning all domains of life. #Comparative genomics # Synthetic Biology



Our research focus on single-cell and spatial biology

Single-cell database

3CA Curated Cancer Cell Atlas — Collected, annotated and analyzed cancer scRNA-seq datasets
- Pan cancer
- 2.3 M cells

Large-scale NSCLC single-cell atlas — 29 datasets, 556 samples, 1,280,000 cells; High-resolution TME; desert, B cell, M cell, T cell; Patient stratification
- Lung cancer
- 1.3 M cells

HLCA extended
- Lung cells
- 2.4 M cells

CZ CELL×GENE DISCOVER
- Human cells
- 59.4 M

HUMAN CELL ATLAS
- Human cells
- 50.1 M

Spatial database database

DB
- All species
- 61.3 M spots

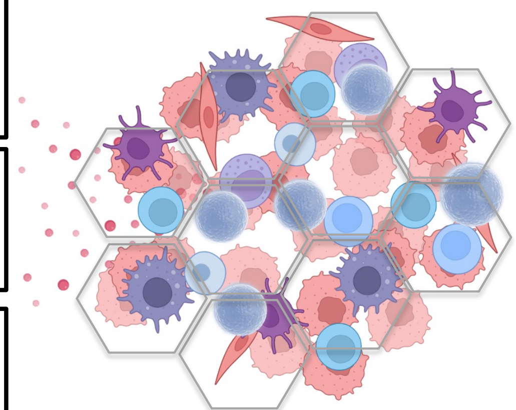
STOmicsDB
- 128 tissues
- 221 datasets
- 17 species

SOAR
- 2,785 samples
- 304 datasets
- 40 tissues

ENABLE MEDICINE
- 20k+ samples
- 20+ disease
- 200+ proteins

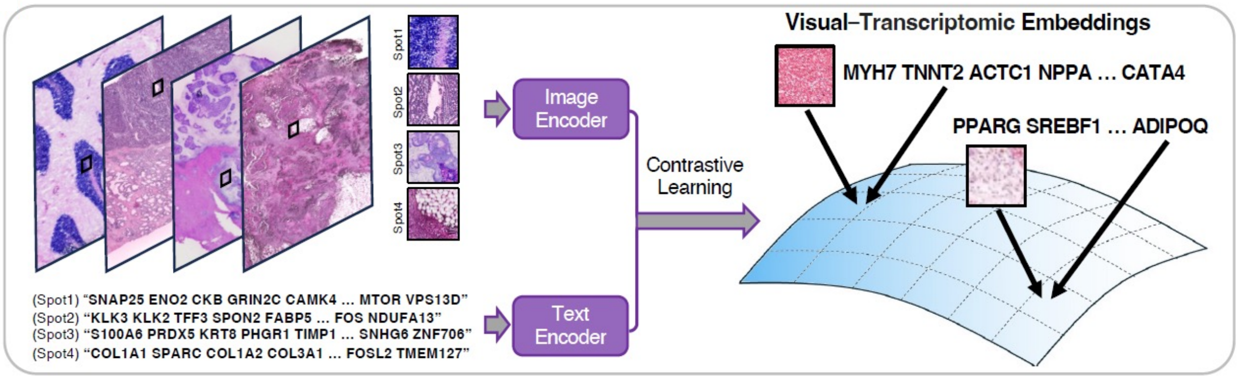
AQUILA
- 6.5K ROIs
- 15.7 M Spots
- 30 disease

SenNet — **HuBMAP** Human BioMolecular Atlas Program — **HTAN** HUMAN TUMOR ATLAS NETWORK

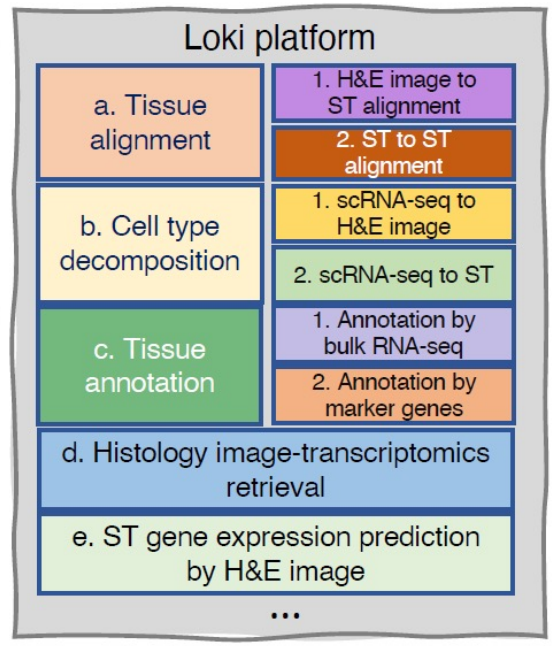


Three "V" define big data:
- High Volume
- High Velocity
- High Variety

LOKI: A visual–omics foundation model to bridge histopathology with spatial transcriptomics



- The first visual-omics foundation model that pairs **H&E histopathology** with **spatial transcriptomics** at patch resolution
- **2.2 million image–transcriptome** pairs from 1,007 samples across 32 organs/disease states
- Converts transcriptomic data into "**sentences**" of top-expressed genes



Chen, W., Zhang, P., Tran, T.N. *et al.* A visual–omics foundation model to bridge histopathology with spatial transcriptomics. ***Nat Methods*** 22, 1568–1582 (2025). https://doi.org/10.1038/s41592-025-02707-1

Guangyu Wang, PhD
Associate Prof,
Methodist

LOKI applications in cancer research

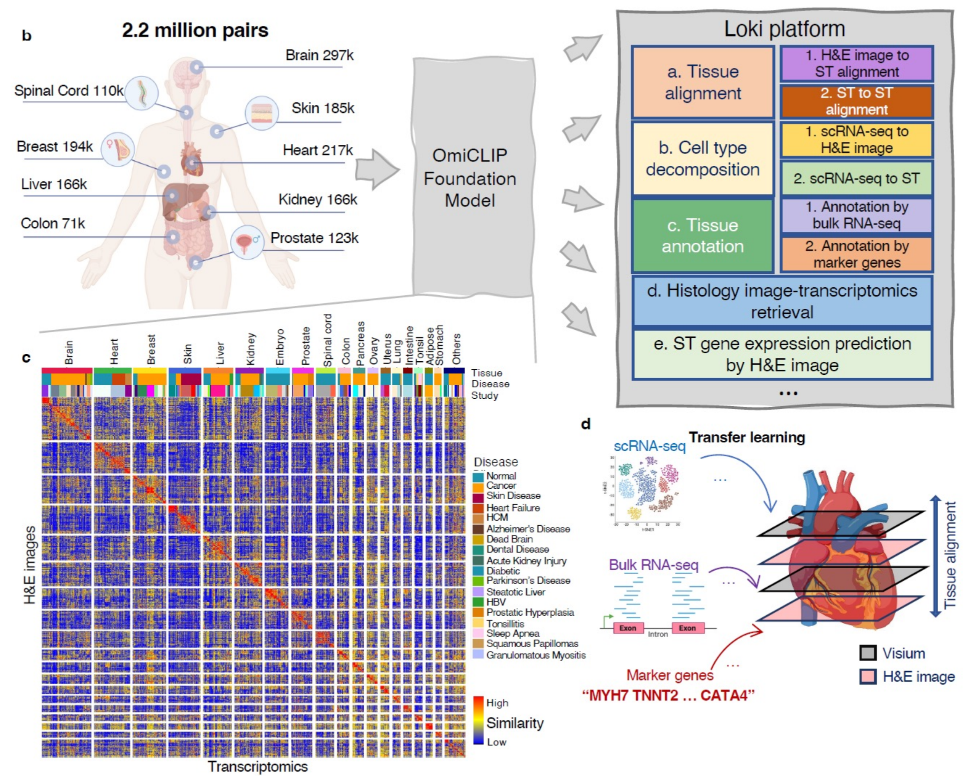
1. **Tissue Alignment** for understanding tumor heterogeneity and invasion patterns

2. **Tumor Annotation** for improving diagnostic accuracy in colorectal and breast cancer samples.

3. **Cancer Cell-Type Analysis** for enabling detailed analysis of epithelial, immune, and stromal cell interactions in triple-negative breast cancer.

4. **Cancer Transcriptomics Retrieval** for matching specific cancer histology patterns in precision oncology.

5. **3D Gene Expression Prediction** for offering cost-effective alternatives to extensive sequencing in cancer studies.



Let us trace the rise of biological foundation models

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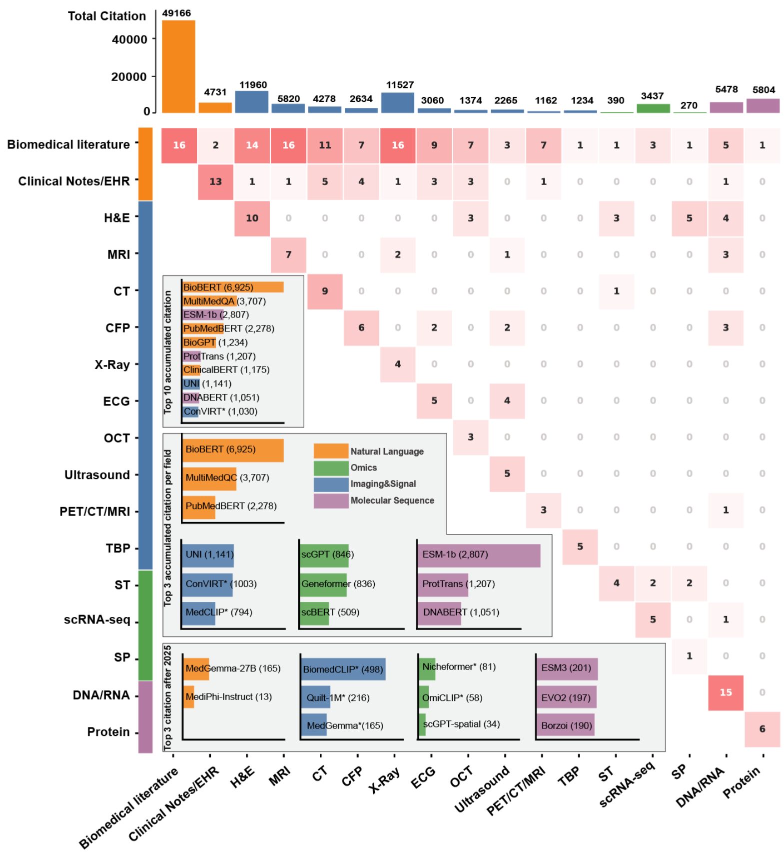
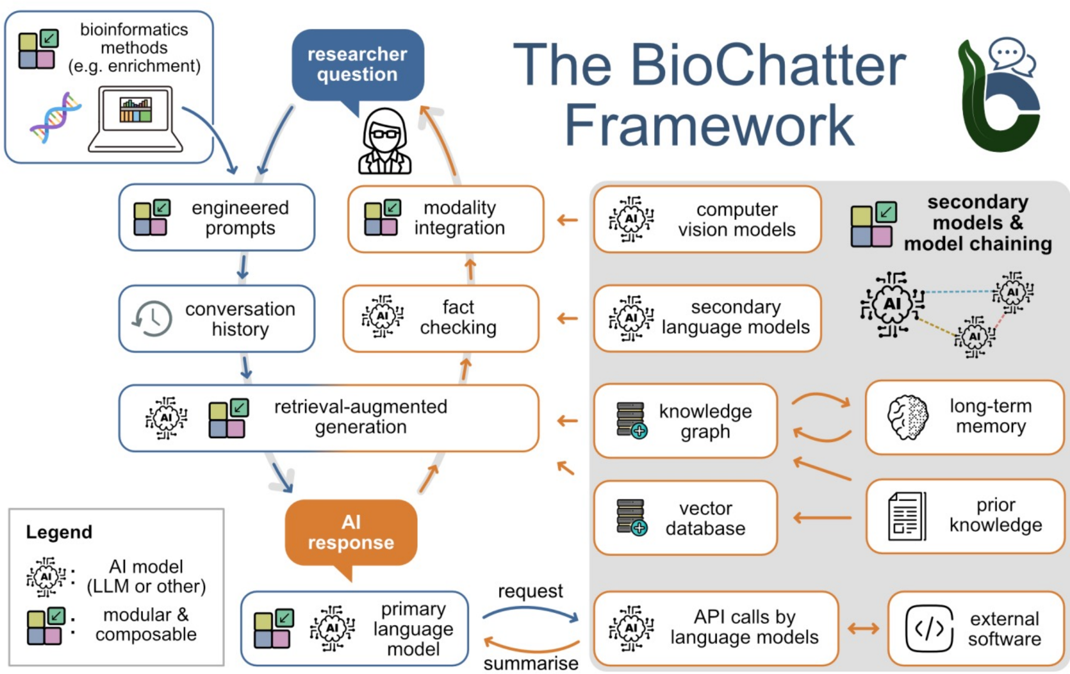


Chang, Y., Cheng, H., Modi, M., *Wang, G., Xu, D., Ma, Q.* Tracing the rise of biomedical foundation models. ***Nat Biotechnol*** (2026). https://doi.org/10.1038/s41587-026-03135-y

- **209** unique foundation models across 16 biomedical data modalities, before 01/31/2026.
- A clear shift towards multimodal modeling, with 123 models (58.9%) integrating 2+ data types, compared to 86 unimodal models (41.1%).
- Mostly designed as encoder-centric with (56.0%), while encoder-decoder architectures remain relatively rare (28.7%), emphasis is on scalable and transferable learning over biological datasets.
- A majority of models (52.6%) are validated on classification tasks, far fewer are evaluated on diagnostic reports (<10%), tissue segmentation (disease or complex tissue structure, <10%), or biological question answering (< 10%).
- **These findings reveal an urgent need for multi-source, human-in-loop, biologically meaningful validation.**

BioChatter: An LLM platform that lets biologists chat directly with their data



Significance

- Open source, adaptable, conversational interface
- **Knowledge graph** integration for false outputs ("hallucinations") of large language models (LLMs)
- Use retrieval-augmented-generation to upload your own documents and get the information you're looking for
- Expandable to support AI-driven insights, causal reasoning, and emerging technologies

https://biochatter.org/
https://bmblx.bmi.osumc.edu/biochatter-next

Lobentanzer, S., Feng, S., Bruderer, N. *et al.* A platform for the biomedical application of large language models. ***Nat Biotechnol*** 43, 166–169 (2025). https://doi.org/10.1038/s41587-024-02534-3

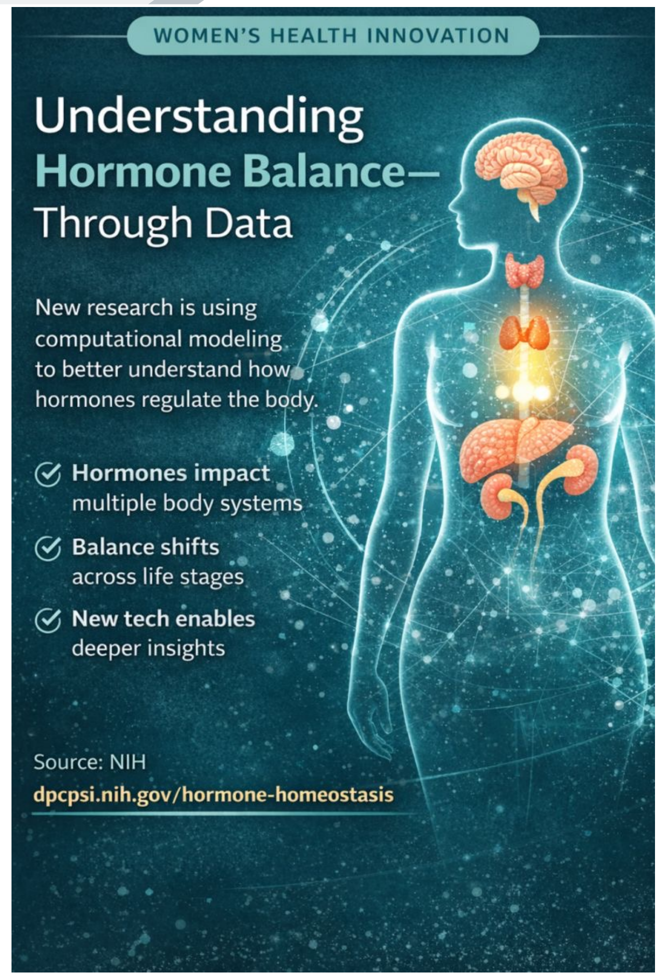
Julio Saez-Rodriguez, PhD
Heidelberg University,
EBI, research head
Germany

Foundation models capture associations and correlations at scale,
yet fall short of true causal inference

NIH Computational Modeling of Hormone Homeostasis (OT2) (April 2026)

The overall goal is to advance the development of in silico computational approaches that model and simulate hormone signaling pathways and homeostasis, with an emphasis on applicability to sex-specific efficacy and toxicity evaluation of therapeutics.

- Kinetic and dynamic modeling;
- Digital twins;
- Systems level biological modeling;
- Multi-scale models; and
- Data-driven predictive models
- 5M per year



ARPA-H: Critical Illness Immunological Reprogramming and Control Point Learning Engine

- **The problem:** ICU immune dysregulation drives preventable deaths
- **The vision:** AI-guided predictive critical care at bedside
- **Digital twin core:** Patient-specific immune system models predicting trajectories
- **Digital twin enablers:** Real-time multi-modal data continuously updating twins
- **Program:** 25-30M, ARPA-H CIRCLE, integrated teams, commercialization built-in

Acknowledgement

All BMBL lab members!
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NSF

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